# EE-482: Network Security and Cryptography

# LECTURE NO 13:

## *i. Digital Signature Standard (DSS)*
## *ii. Compression*
## *iii. Decompression*

**Engr. Tehseen Ahsan**
**Assistant Professor, Computer Engineering Department**
**HITEC University Taxila Cantt, Pakistan**

# Organization of Lecture

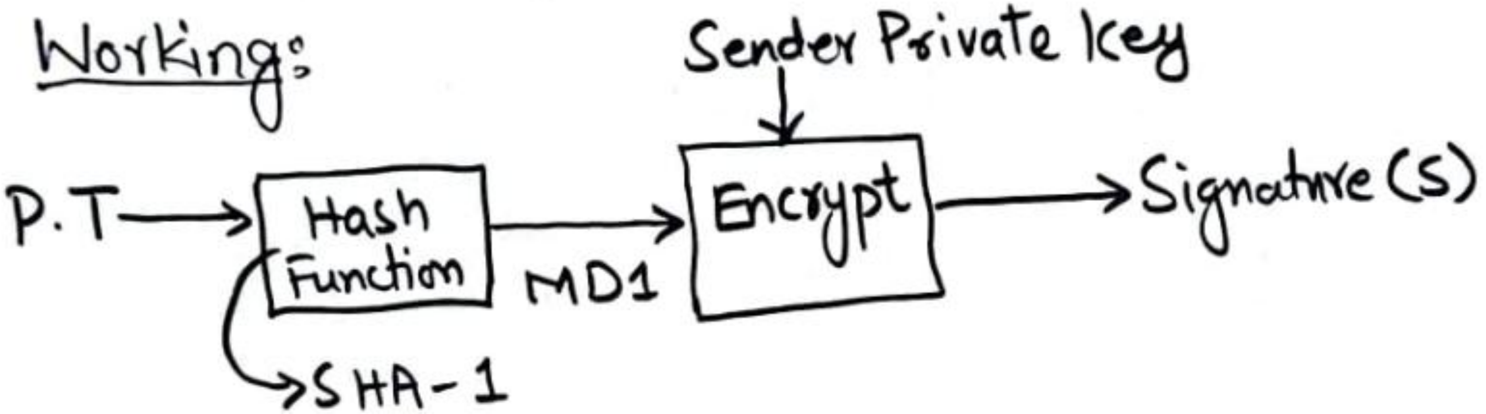
1. Digital Signature Standard (DSS)
2. Compression
3. Decompression

# 1. Digital Signature Standard (1/7)

## 1) Digital Signature Standard (DSS)

- Developed by National Institute of standards & Technology (NIST) in 1991.
- Uses an algorithm called Digital Signature Algorithm (DSA).
- DSA is based on asymmetric key cryptography.

Working:

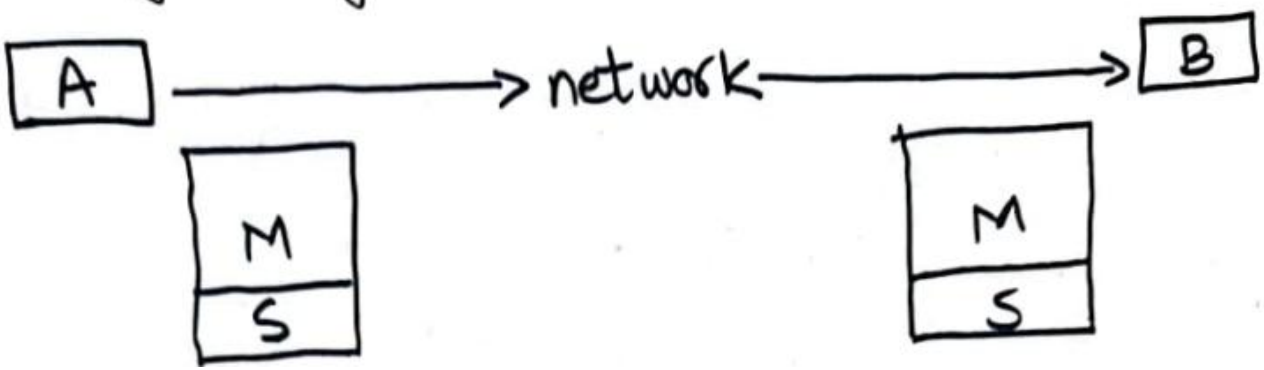


# 1. Digital Signature Standard (2/7)

Step 1: Sender (A) uses SHA-1 message digest algorithm to calculate MD1 over the original message (plain text).

Step 2: Sender (A) encrypts the MD1 with its private key. Output of this process is the Digital Signature of A.

Step3: Sender (A) sends original message along with digital signature (S) to receiver (B).

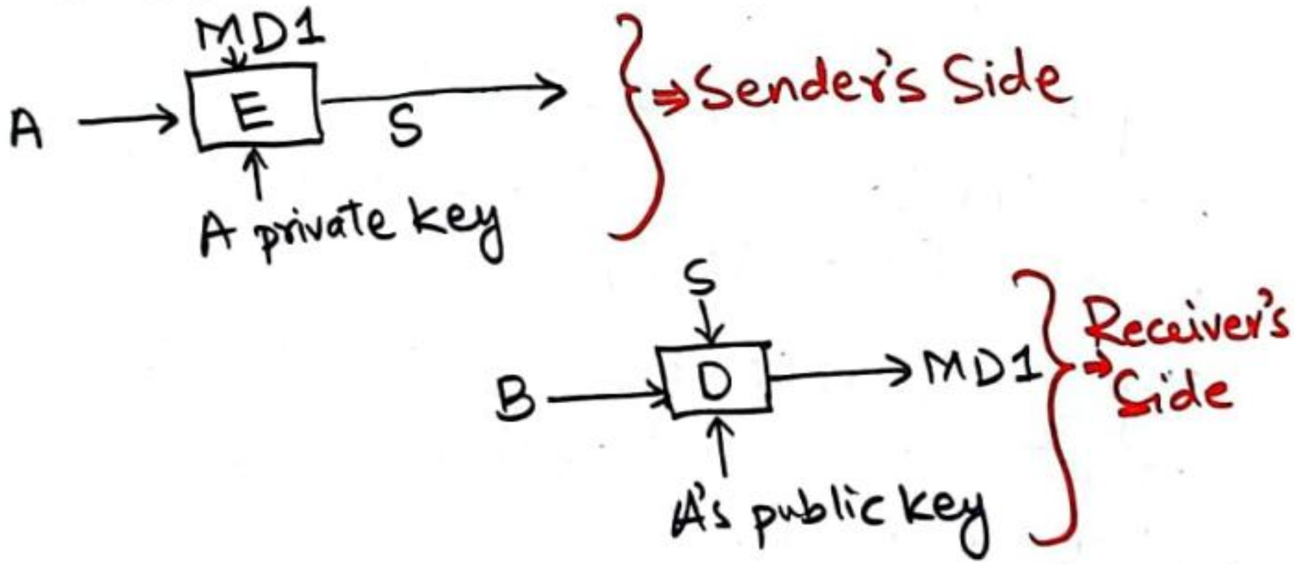


# 1. Digital Signature Standard (3/7)

Step 4: Receiver (B) receives the original message (M) & sender's (A) digital signature (S).

B uses same message digest algorithm as used by A to calculate its own message digest (MD2).

Step 5: B uses A's public key to decrypt digital signature (S).




Output of decryption is the message digest MD1.

# 1. Digital Signature Standard (4/7)

Step 6: B compare the two message digests

MD2 ⟶ from step 4

MD1 ⟶ retrieved from A's digital signature

If MD1 = MD2

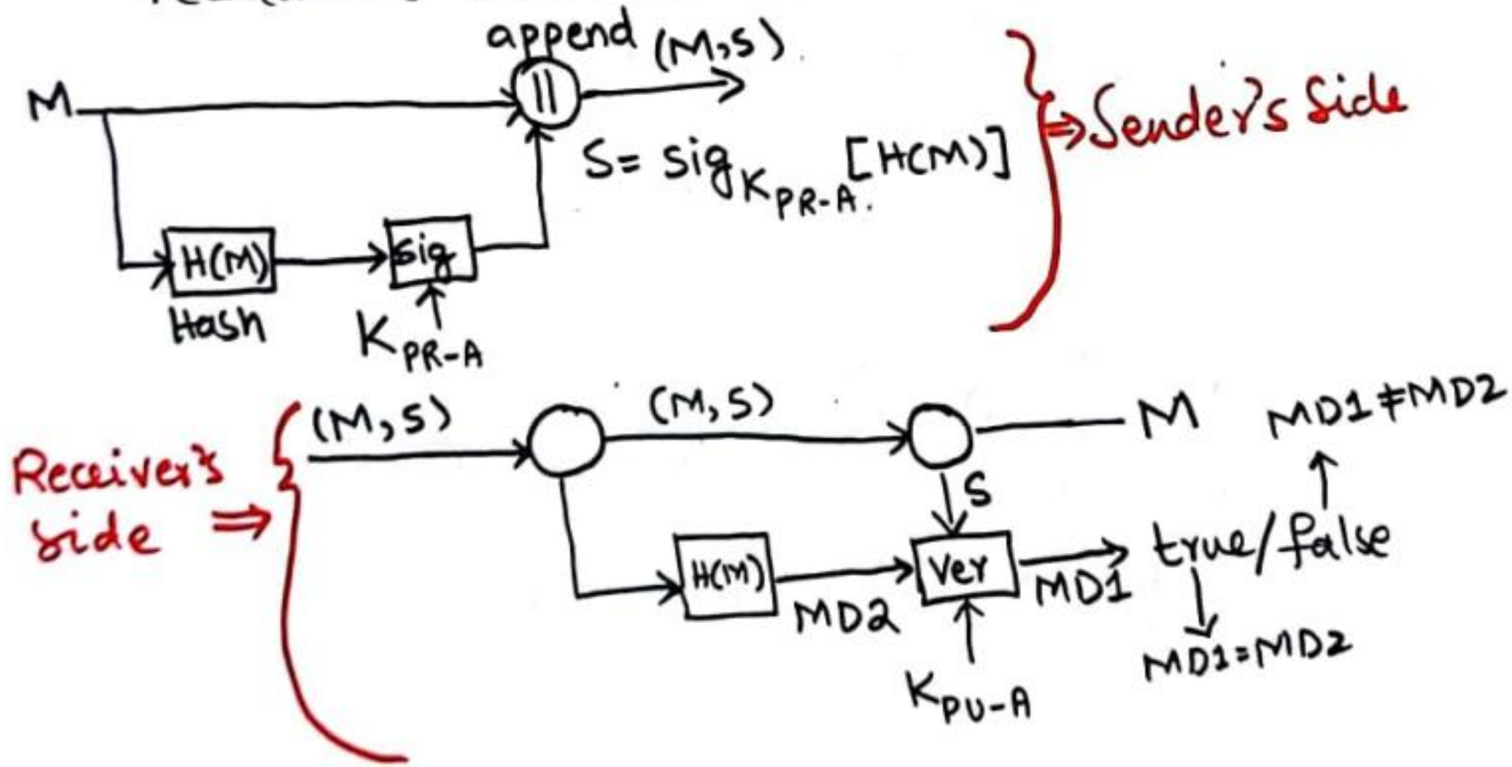
→ B accepts the original message (M) as correct and unaltered.

→ B is also assured that message came from A & not from someone posing as A.

# 1. Digital Signature Standard (5/7)

Advantages of using private/public key Scheme:

1) Since only A knows about A's private key, the attacker cannot use A's private key to encrypt the message digest.
2) Attacker can't sign the MD even if he is able to recalculate the MD.



# 1. Digital Signature Standard (6/7)

**Arbitrated Digital Signatures:**

↳ What if sender A annonymously publishes his private key and then A falsely claims his private key was stolen & someone else forged his signature.

↳ Best if an arbitrator is present for resolving such disputes.

↳ Every message from A to B goes through a trusted party 'X'.

↳ X performs a no. of tests to verify the message content and its origin.

↳ Message is then time stamped & sent to B.

# 1. Digital Signature Standard (7/7)

**Handshaking b/w A and X:**

$$M \| E_{K_X}[ID_X \| H(M)]$$

$E_{K_X}$ : encrypted with a key (private key) assigned by arbitrator.

Public key ⟶ sent to B.

Private key ⟶ known to A & X.

$ID_X$ : Identification of X.

$H(M)$ : hash value / message digest of message M.

$\|$ : concatenation or Append.

# 2. Compression (1/4)

* ZIP:

↳ used for emails.

↳ PGP: Pretty good privacy.

↳ Written for UNIX environment

↳ Different versions

- WINRAR
- PKZIP
- WINZIP
- ZIPIT

↳ exploit the fact that words & phrases within a text stream are likely to be repeated.

↳ When a repetition occurs, the repeated sequence can be replaced by a short code.

# 2. Compression (2/4)

Example:

THE BROWN FOX[13] JUMPED[18] OVER THE BROWN FOXY[39] JUMPING[45] FROG[53]

Counting Blanks too ⇒ 53 octets (53x8=424).
(424 bits long)

↳ The algorithm looks for repeated sequences.

↳ When a repetition is encountered, the algorithm continues scanning till repetition ends.

↳ The repeated sequence is replaced by a code.
  ↳ pointer to prior sequence
  ↳ length of sequence.

First repeated sequence: THE BROWN FOX
pointer: 26 (distance to previous)
(39−13)
length: 13 (length of repetition)

# 2. Compression (3/4)

↳ 2 options for encoding

| Option 1 | Option 2 |
|---|---|
| 8 bit pointer<br>4 bit length | 12 bit pointer<br>6 bit length |
| Short compression | Larger Sequence |

⇒ 2 bit header indicates which option is chosen

00 indicates option 1
01 indicates option 2

→ The second occurance of THE BROWN FOX encoded as

$\langle 00_b \rangle \langle 26_d \rangle \langle 13_d \rangle$

00 00011010 1101

# 2. Compression (4/4)

↳ Second repeated sequence is

JUMP

$\rightarrow$ (45-18)

Pointer to previous 27

length 5

Encoded as

$\langle 0\ 0_b \rangle$ $\langle 27_d \rangle$ $\langle 5_d \rangle$

00 00011011 0101

THE BROWN FOX JUMPED OVER THE BROWN FOXY JUMPING FROG

THE BROWN FOX JUMPED OVER $0_b 26_d 13_d$ Y $0_b 27_d 5_d$ ING FROG.

# 3. Decompression

3) **Decompression Algorithm:** When an encoded string is encountered, the decompression algorithm uses <pointer> and <length> fields to replace the code with actual text string.

**Compression Ratio:**

Uncompressed = 53 octets = $53 \times 8 = 424$ bits

Compressed = 36 octets = $36 \times 8 = 288$ bits

Codes = $2 \times 14 = 28$ bits

$$\text{Ratio} = \frac{[\text{Compressed} + \text{Codes}]}{\text{Uncompressed}} \times 100$$

$$= \frac{[288 + 28]}{424} \times 100$$

$$\text{Ratio} = \frac{316}{424} \times 100 = 74.52\%$$

# References: Textbooks

1. *"Cryptography and Network Security", by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *"Cryptography and Network Security Principles and Practice", by William stallings, 6th Edition, Pearson.*
3. *"Computer Security Principles and Practice", by William stallings, Lawrie Brown, 4th Edition, Pearson.*